Daniel **Saromo-Mori**

MECHATRONIC ENGINEER · INVENTOR OF THE AUTO-ROTATING NEURAL NETWORKS (ARNN) · AI RESEARCHER AND LECTURER

Portfolio: www.danielsaromo.xyz | DanielSaromo | danielsaromo | Daniel Saromo-Mori

About me

→ I'm Daniel, a mechatronics engineer passionate about AI-powered robot control. My main research interest is *robotics × machine learning: robot learning*. As a result of my research in AI, I have invented the ARP and the ARNN —algorithms that I have presented in six countries. Besides, I have **6+ years of experience** in research and teaching AI, ML, and Data Science. Also, I have **practical experience in Robot Learning**, like my AI-guided spider robot, work recognized in the United States with the Innovation Award at IMECE 2019.

→ ***Language Skills:*** English (proficient level), French (elementary level), Italian (advanced level), and Spanish (native).

Professional Experience

Siemens — *Prague, Czechia*

APPLICATION ENGINEER FOR IT/OT INTEGRATION (WORKING STUDENT / DPČ) — *Since Aug. 2025*

Education

M.Sc. in Cybernetics and Robotics, with an Specialization in AI · Currently enrolled — *Prague, Czechia*

CZECH TECHNICAL UNIVERSITY IN PRAGUE (1ST IN CZECHIA: QS ENGINEERING RANKING 2025) — *Since Sep. 2024*

- I have been awarded a **merit scholarship** for excellent study results achieved in the WS of the AY 2024/2025.

M.Sc. in Automation and Control Engineering · Weighted mark after 3 semesters: 25.6/30 — *Milan, Italy*

POLITECNICO DI MILANO (13TH IN THE WORLD: QS ENGINEERING RANKING 2022) — *Sep. 2022 - Feb. 2024*

B.Sc. in Mechatronics Engineering & Mechatronics Engineering Professional Degree — *Lima, Peru*

PONTIFICIA UNIVERSIDAD CATÓLICA DEL PERÚ (1ST IN PERU: QS RANKING 2025) — *03/2014 - 07/2019, 08/2019 - 11/2020*

- **Bachelor's average course grade:** 15.70 (Scale: minimum: 0, required to pass: 11, maximum: 20).
- **Academic ranking:** Top fifth of class (6th of 32 mechatronics graduates) · Top 6.66% of the students of the Faculty of Science and Engineering.
- **Theses title:** Intelligent spider robot for detecting anti-personnel metallic landmines in uneven terrain.
- **Professional Degree Thesis Awards:** - Extraordinary Support Funding for Undergraduate Research Thesis.
  - **Ranking:** Degree thesis unanimously awarded by the tribunal with the qualification of *outstanding*.
- **B.Sc. Thesis Awards:** - Best bachelor's thesis and poster presentation at the PUCP Mechatronics Workshop of the semester 2019-1.
  - Innovation Recognition Award at the International Mechanical Engineering Congress & Exposition (**IMECE**) 2019.
- **Theses advisors:** Dr. Elizabeth Villota and Dr. Edwin Villanueva.

Scientific Publications · Currently with more than 150 citations reported in Google Scholar

[C4] Valdenegro-Toro, M. and **Saromo, D.** "A Deeper Look into Aleatoric and Epistemic Uncertainty Disentanglement", *LXCV Workshop at CVPR 2022*. Louisiana, U.S.A. 2022 · Paper presented in the poster session and was one of the few selected for an oral presentation.

[C3] Bravo, L., **Saromo, D.**, and Villota, E. "Smart Insole Sensor for vGRF Measurement", *9th International Symposium on Sensor Science*. Warsaw, Poland. 2022.

[C2] **Saromo, D.**, Bravo, L., and Villota, E. "Smart Sensor Calibration with Auto-Rotating Perceptrons", *LXAI Workshop at ICML 2020*. Vienna, Austria. 2020 · Paper presented in the poster session and was one of the few selected for an oral presentation.

[C1] **Saromo, D.**, Villota, E., and Villanueva, E. "Auto-Rotating Perceptrons", *LXAI Workshop at NeurIPS 2019*. Vancouver, Canada. 2019 · Paper presented in the poster session and was one of the few selected for an oral presentation.

[T1] **Saromo, D.** "Intelligent spider robot for detecting anti-personnel metallic landmines in uneven terrain", *Pontificia Universidad Católica del Perú*. Lima, Peru. 2020 · Thesis published in Spanish. English abstract available: link.

Teaching Experience

- PUCP's Continuing Education Department · Teacher at Specialization Diplomas — *Lima, Peru*

LECTURER
- Diploma in Development of AI Applications (**Course:** AI for Games): 2019-2, 2020-1, 2020-2, 2021-1, 2021-2, 2022-1, 2022-2, 2023-1, 2023-2, 2024-1, 2024-2, 2025-1. — *Since Sep. 2019*
- Diploma in Data Analytics (**Course:** Data Analysis Methods for Time Series): 2022-1. — *Jun. 2022 - Oct. 2022*

- PUCP's Center for Advanced Manufacturing Technologies (CETAM) — *Lima, Peru*

LECTURER
Courses:
- ML for Industry (2020-2, 2021-1, 2021-2, 2022-1, 2022-2, 2023-1, 2023-2); — *Sep. 2020 - Sep. 2023*
- Python for Data Science (2021-1, 2021-2, 2022-1, 2022-2, 2023-1). — *Jun. 2021 - Sep. 2023*

- National Meteorological and Hydrological Services (SENAMHI) · Peruvian Government Entity — *Lima, Peru*

LECTURER — **Course:** Introduc. to AI and ML for National Meteorological and Hydrological Services. — *May. 2022 - Jun. 2022*

- PUCP Undergraduate School · Faculty of Science and Engineering — *Lima, Peru*

TEACHING ASSISTANT — **Undergrad. courses:** AI (2019-1), ML (2019-2), Computer Science Applications (2019-2). — *Mar. 2019 - Dec. 2019*

Honors & Awards

- 2023 **1st place**, at the Pitch Competition 2023 organized by Entrepreneurship Club POLIMI — *Milan, Italy*
- 2019 **LXAI Travel Grant**, for attending NeurIPS 2019 to be an oral and poster presenter · 1860 USD — *Vancouver, Canada*
- 2019 **Innovation Recognition Award**, Old Guard 63rd Anual Oral Competition (World Finals at IMECE) · 250 USD — *Utah, U.S.A.*
- 2019 **ASME Travel Award**, to represent PUCP and South America at ASME IMECE Finals Competition · 1500 USD — *Utah, U.S.A.*
- 2019 **1st place + Technical Award**, Old Guard Oral Presentation Competition (ASME E-FEST South America) · 850 USD — *Lima, Peru*

Professional and Research Experience

German Research Center for Artificial Intelligence (DFKI) — *Bremen, Germany*

GUEST RESEARCHER · REMOTE MODE — *Aug. 2020 - Dec. 2020*

- **Auto-Rotating Neural Networks (ARNN):** I extended the ARP concept and created a new neural model family named Auto-Rotating Neural Networks. I've implemented dense, recurrent, LSTM, GRU, and convolutional layers with the Auto-Rotating operation; and obtained promising results. *Research advisor:* Dr. Matias Valdenegro-Toro.
- We are testing the implementation of the ARNN, to validate and compare their performance against equivalent models without the Auto-Rotation. Experiments ran in the research center's GPU clusters. Results presented at the Online Asian Machine Learning School (OAMLS).

PUCP Applied Robotics and Biomechanics Research Group (GIRAB) — *Lima, Peru*

RESEARCH ASSISTANT — *Mar. 2020 - July 2020*

- **Smart Sensor Calibration with Auto-Rotating Perceptrons:** In this paper, we applied the ARP to calibrate a wearable force sensor. By changing classic neurons to ARP, we obtained 15x better performance of the neural networks (i.e., lower loss). *Research advisor:* Dr. Elizabeth Villota.

Talks & Presentations

- Oct. 2024 **Pitch: Auto-Rotating Neural Networks**, Dny AI: AI-focused series of events taking place across Czechia — *Prague, Czechia*
- June 2022 **Paper exposition: A Deeper Look into Aleat. and Epist. Uncert. Disentang.**, Speaker at LXCV CVPR — *New Orleans, U.S.A.*
- Nov. 2021 **Poster presentation: Auto-Rotating Neural Networks**, Online Asian Machine Learning School at ACML — *Singapore, Singapore*
- Jul. 2020 **Paper exposition: Smart Sensor Calibration with Auto-Rotating Perceptrons**, Speaker at LXAI ICML — *Vienna, Austria*
- Dec. 2019 **Paper exposition: Auto-Rotating Perceptrons (ARP)**, Speaker at LXAI NeurIPS — *Vancouver, Canada*

Continuing Education

- Jun. 2025 **7th International Summer School of AI and Games**, by Georgios Yannakakis and Julian Togelius — *gameaibook.org*
- Aug. 2024 **Robot Operating System (ROS) (Grade: 20/20)**, PUCP Center for Engineering Vinculation (FABRICUM) — *FABRICUM PUCP*
- Nov. 2023 **Disaster Risk Monitoring Using Satellite Imagery**, NVIDIA Deep Learning Institute — *NVIDIA DLI*
- Aug. 2022 **Oxford Machine Learning Summer School**, Oxford University & AI for Global Goals — *Oxford University*
- Jun. 2022 **AutoCAD - Level: Intermediate**, National University of Engineering (UNI) — *UNI*
- Mar. 2022 **Solidworks - Level: Intermediate**, National University of Engineering (UNI) — *UNI*
- Jan. 2022 **Solidworks - Level: Basic**, National University of Engineering (UNI) — *UNI*
- Nov. 2021 **Online Asian Machine Learning School**, Asian Conference on Machine Learning (ACML) — *ACML 2021*
- Jul. 2021 **Robot Operating System (ROS)**, Center for Advanced Manufacturing Technologies (CETAM) — *CETAM PUCP*
- Nov. 2020 **Scrum Master Certification Training**, IEEE Ricardo Palma University Student Branch — *IEEE Peru Section*
- Jul. 2019 **Getting started with AI on Jetson Nano**, NVIDIA Deep Learning Institute — *NVIDIA DLI*
- Nov. 2018 **PyTorch Scholarship Challenge**, Udacity / Facebook — *Udacity / Facebook*
- Apr. 2018 **Machine Learning for Data Science and Analytics**, Columbia University — *edX*
- Feb. 2017 **PCB design with international standards oriented to manufacturing**, AlDelta Technologies — *AlDeltaTec.com*
- May 2016 **Embedded Systems – Shape the World**, University of Texas at Austin — *edX*
- May 2015 **Introduction to Robotics**, Queensland University of Technology — *QUT MOOC*

Skills

Technical tools — **Robotics:** ROS-1. | **Automation:** TIA Portal (PLC/HMI). LabView. | **Web:** HTML, CSS, Jekyll. | **CAD/CAE:** *Mechanics:* Inventor, SolidWorks, AutoCAD. *Electronics:* EagleCAD, Altium Designer, Circuit Maker, Proteus, B2 Spice, NI Multisim. | **Embedded Systems:** ATmega328P, ATmega2560 (Arduino IDE); Raspberry Pi (SBC and RP2040); Jetson Nano. | **Coding:** Python, MicroPython, C, C++, MATLAB, VBA (for Excel), LaTeX. | **Frameworks:** Linux, Git, Tensorflow, Keras, Scikit-Learn, PyTorch, NumPy, Pandas, Seaborn, Matplotlib, Plotly, OpenCV, OpenAI Gym, PyDuino Bridge.

Technical skills — **Mechanics:** Parametric 3D modeling and assembly. Technical drawing. **Electronics:** Schematic drawing. PCB design, assembly, and testing. Excellent soldering skills (THT and SMT). | **Automatic Control:** Classical and state-space. | **Artificial Intelligence and Machine Learning:** Deep Learning (MLP, CNN, ARNN, ARP). Search algorithms and heuristics. Bio-inspired optimization (ABC, ACO, and PSO). Decision Trees. Random Forests. SVM. Clustering. PCA. Ensemble Learning. Transfer Learning. Deep Reinforcement Learning (DQN, DDQN). Computer Vision. Image Segmentation. Image Style Transfer.

Soft skills — Strong abilities in public speaking, teaching, teamwork, and leadership. Maker spirit. Curiosity and perseverance.